Per-site ΔRMSE from adding disturbance — M3 (C+T) vs M4 (C+T+D)

Below 0 = D reduces the site's RMSE | bars coloured by relative mortality class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (93 sites, 372 site-years) · deadtrees v2-2

XGBoost (Optuna) · 12m | **XGBoost (Optuna) · 24m** | **LightGBM (Optuna) · 12m** | **LightGBM (Optuna) · 24m**

GPP (g C m⁻² season⁻¹)
59% improved | 66% improved | 68% improved | 61% improved

NEP (g C m⁻² season⁻¹)
62% improved | 60% improved | 66% improved | 65% improved

ET (mm season⁻¹)
49% improved | 47% improved | 49% improved | 52% improved

WUE (g C mm⁻¹)
60% improved | 59% improved | 58% improved | 55% improved

ΔRMSE = RMSE(with D) − RMSE(without D)

Sites (sorted by ΔRMSE within each panel)

Relative mortality
site maximum, 500 m
(breaks: mean ± 0.5 SD)

- Low (< 18.3 %, 35 sites)
- Mid (18.3–30.0 %, 32 sites)
- High (> 30.0 %, 26 sites)